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of individual squares n, the process ends.

p, c : = 1 , 0

while p ≠ n

p, c := p + 1, c+1

We have observed (in Example 8.2) that p - c is an invariant of the assignment p, c := p + 1, c + 1. Let p - c = k, where k is a constant. The points in the algorithm where p - c = k is true are shown in the algorithm below, and in the flowchart of Figure 8.2.

p, c : = 1 , 0

1. -- p - c = k

while p ≠ n

2. -- p - c = k

p, c := p+1, c+1

3. -- p - c = k

4. --p-c=k,p=n

The loop invariant p- c = k is True at the start of the loop (line 1). Moreover, at the start of the loop, p- c = 1. Therefore, k = 1, and the loop invariant is p - c = 1

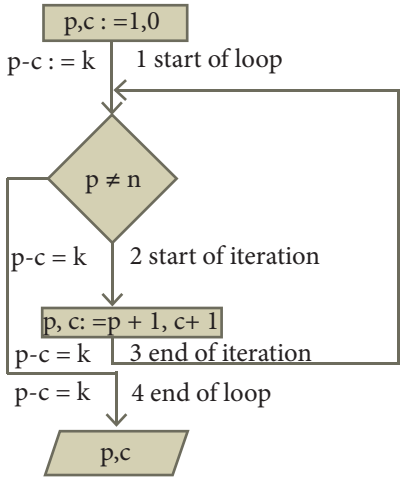

Figure 8.2: The points where the loop invariant is true

When the loop ends (line 4), the loop invariant is still true (p - c = 1). Moreover, the loop condition is false (p = n). From p - c = 1 and p = n,

1. p — c = 1 loop invariant

2. p = n end of the loop

3. n — c = 1 from 1, 2

4. c = n — 1 from 3

When the process ends, the number of cuts is one less than the number of squares.

Example 8.7. There are 6 equally spaced trees and 6 sparrows sitting on these trees,one sparrow on each tree. If a sparrow flies from one tree to another, then at the same time, another sparrow flies from its tree to some other tree the same distance away, but in the opposite direction. Is it possible for all the sparrows to gather on one tree?